

Predicate Logic

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GROUP 314
OPTION 2

- ① H_1 : Every member ^{that} is a veteran and active ~~is~~ ^{they are} eligible for the board.
 H_2 : Any member ^{that} is eligible for the board ~~they~~ must attend the annual gala.
 H_3 : Every member ^{that} is a veteran but not active ~~they~~ is an honorary member.
 H_4 : Sarah is a veteran.
 H_5 : Sarah is not an honorary member.
 e : Sarah must attend the annual gala.

• D = the set of all members, Sarah $\in D$ (constant)

• Predicate Symbols:

$v(x) : D \rightarrow \{T, F\}$, $v(x) = T$ if x is a veteran

$a(x) : D \rightarrow \{T, F\}$, $a(x) = T$ if x is active

$e(x) : D \rightarrow \{T, F\}$, $e(x) = T$ if x is eligible for the board

$g(x) : D \rightarrow \{T, F\}$, $g(x) = T$ if x must attend the annual gala

$R(x) : D \rightarrow \{T, F\}$, $R(x) = T$ if x is an honorary member

$$H_1: (\forall x) [(v(x) \wedge a(x)) \rightarrow e(x)]$$

$$H_2: (\forall x) (e(x) \rightarrow g(x))$$

$$H_3: (\forall x) [(v(x) \wedge \neg a(x)) \rightarrow R(x)]$$

$$H_4: v(\text{Sarah})$$

$$H_5: \neg R(\text{Sarah})$$

$$e: g(\text{Sarah})$$

$$H_1 \wedge H_2 \wedge H_3 \wedge H_4 \wedge H_5 \wedge \neg C(1) \quad \checkmark$$

$$I \propto \text{for } (1)$$

$$H_1: (\forall x) ((\neg r(x) \wedge a(x)) \rightarrow e(x)) \quad (2) \quad \checkmark$$

$$H_2: (\forall x) (e(x) \rightarrow g(x)) \quad (3) \quad \checkmark$$

$$H_3: (\forall x) ((\neg r(x) \wedge \neg a(x)) \rightarrow R(x)) \quad (4) \quad \checkmark$$

$$H_4: \neg r(\text{Sarah})$$

$$H_5: \neg R(\text{Sarah})$$

$$\neg C: \neg g(\text{Sarah})$$

$\neg C$ for (2), Sarah - used for instantiation

$$(\neg r(\text{Sarah}) \wedge a(\text{Sarah})) \rightarrow e(\text{Sarah}) \quad (5)$$

copy (2)

$$(5) \neg r(\text{Sarah}) \vee \neg a(\text{Sarah}) \quad \beta \text{ for } (5)$$

$$\neg r(\text{Sarah}) \quad \neg a(\text{Sarah}) \quad \beta \text{ for } (6)$$

Sarah used for instantiation

$$e(\text{Sarah}) \rightarrow g(\text{Sarah}) \quad (4) \quad \text{Sarah - used for instantiation}$$

$$e(\text{Sarah}) \rightarrow g(\text{Sarah}) \quad (4)$$

copy (3)

$$\neg e(\text{Sarah}) \quad g(\text{Sarah}) \quad \beta \text{ for } (7)$$

$$(\neg r(\text{Sarah}) \wedge \neg a(\text{Sarah})) \rightarrow R(\text{Sarah}) \quad (8)$$

copy (4)

$$(8) \neg r(\text{Sarah}) \vee \neg a(\text{Sarah}) \quad \beta \text{ for } (8) \quad R(\text{Sarah})$$

$$\neg r(\text{Sarah}) \quad \neg a(\text{Sarah}) \quad \beta \text{ for } (9)$$

$H_1 \wedge H_2 \wedge H_3 \wedge H_4 \wedge H_5 \wedge \neg C$ has a closed semantic tableau $\Rightarrow$
 $\Rightarrow$ the conclusion C is deducible from the set
of hypothesis $\{H_1, H_2, H_3, H_4, H_5\}$

② H₁: Every rebel who possesses a mockingjay pin is recognized as a leader.

H₂: Anyone who is a leader and issues a signal is followed by the district.

H₃: Every rebel possesses at least one mockingjay pin.

H₄: Katniss is a rebel who issues a signal.

C: Katniss is followed by the district.

- Domain: the set of all entities (people + objects) = Δ

- Predicate Symbols:

$r(x) : \Delta \rightarrow \{T, F\}$, $r(x) = T$ if x is a rebel

$p(x, y) : \Delta \times \Delta \rightarrow \{T, F\}$, $p(x, y) = T$ if x has mockingjay pin y

$l(x) : \Delta \rightarrow \{T, F\}$, $l(x) = T$ if x is a leader

$i(x) : \Delta \rightarrow \{T, F\}$, $i(x) = T$ if x issues a signal

$f(x) : \Delta \rightarrow \{T, F\}$, $f(x) = T$ if x is followed by the district

- Constants: $K(\text{Katniss}) \in \Delta$

$$H_1: (\forall x)(\forall y)((r(x) \wedge p(x, y)) \rightarrow l(x)) \equiv (\forall x)(\forall y)(\neg r(x) \vee \neg p(x, y) \vee l(x))$$

$$H_2: (\forall x)((l(x) \wedge i(x)) \rightarrow f(x)) \equiv (\forall x)(\neg l(x) \vee \neg i(x) \vee f(x))$$

$$H_3: (\forall x)(r(x) \rightarrow (\exists y)(p(x, y))) \equiv (\forall x)(\neg r(x) \vee (\exists y)(p(x, y)))$$

$$H_4: r(K) \wedge i(K)$$

$$C: f(K)$$

$$\left. \begin{aligned} H_1 P &= (\forall x)(\forall y)(\neg r(x) \vee \neg p(x, y) \vee l(x)) \\ H_2 P &= (\forall x)(\neg l(x) \vee \neg i(x) \vee f(x)) \\ H_3 P &= (\forall x)(\exists y)(\neg r(x) \vee p(x, y)) \end{aligned} \right\} \text{prenex forms}$$

$[y \leftarrow f(x)]$, $f(x)$ - Skolem function

$$H_3^S = (\forall x)(\neg r(x) \vee p(x, f(x)))$$

$$C_1 = (\exists x) \neg \neg(x) \vee (\exists y) \neg p(x, y) \vee (\exists z) \neg l(x)$$

$$C_2 = (\exists x) \neg l(x) \vee (\exists y) \neg i(x) \vee (\exists z) \neg f(x)$$

$$C_3 = (\exists x) \neg \neg(x) \vee (\exists y) \neg p(x, y) \vee (\exists z) \neg f(x)$$

$$C_4 = (\exists x) \neg \neg(K)$$

$$C_5 = (\exists x) \neg i(K)$$

$$C_6 = (\exists x) \neg f(K)$$

$$C_7 = \text{Res}^{\text{lock Pr}} (C_3, C_4) = (\exists x) \neg p(K, y) \vee (\exists z) \neg l(K)$$

$$\theta_1 = [x \leftarrow K]$$

$$C_8 = \text{Res}^{\text{lock Pr}} (C_7, C_4) = (\exists x) \neg p(K, y) \vee (\exists z) \neg l(K)$$

$$\theta_1 = [x \leftarrow K]$$

$$\theta_2 = [y \leftarrow f(K)]$$

$$C_9 = \text{Res}^{\text{lock Pr}} (C_8, C_7) = (\exists x) \neg l(K)$$

$$\theta_2 = [y \leftarrow f(K)]$$

$$C_{10} = \text{Res}^{\text{lock Pr}} (C_9, C_2) = (\exists x) \neg i(K) \vee (\exists z) \neg f(K)$$

$$\theta_1 = [x \leftarrow K]$$

$$C_{11} = \text{Res}^{\text{lock Pr}} (C_{10}, C_5) = (\exists x) \neg f(K)$$

$$C_{12} = \text{Res}^{\text{lock Pr}} (C_{11}, C_6) = \square$$

Conclusion

$$C_1 \wedge C_2 \wedge C_3 \wedge C_4 \wedge C_5 \wedge C_6 \vdash_{\text{lock. Res}}^{\text{Prod}} \square \Rightarrow H_1, H_2, H_3, H_4 \vdash C,$$

so the conclusion is derivable from the premises

$$\{H_1, H_2, H_3, H_4\}$$

③ H_1 : Every plant that is in the sunlight is considered healthy.

H_2 : All vegetables are categorized as plants.

H_3 : If a sensor detects an object that doesn't need water, then the object is in the sunlight.

H_4 : Every vegetable is detected by at least one sensor.

H_5 : The tomato is a vegetable and it does not need water.

c : The tomato is healthy.

- $\Delta = (\text{domain})$ the set of all objects in the greenhouse
 $\text{dom}(\text{tomato}) \in \Delta$ (=constant)

- Predicate symbols:

$p(x) : \Delta \rightarrow \{T, F\}$, $p(x) = T$ if x is a plant

$v(x) : \Delta \rightarrow \{T, F\}$, $v(x) = T$ if x is a vegetable

$\Delta(x) : \Delta \rightarrow \{T, F\}$, $\Delta(x) = T$ if x is in the sunlight

$R(x) : \Delta \rightarrow \{T, F\}$, $R(x) = T$ if x is healthy

$w(x) : \Delta \rightarrow \{T, F\}$, $w(x) = T$ if x needs water

$d(x, y) : \Delta \times \Delta \rightarrow \{T, F\}$, $d(x, y) = T$ if x is detected by sensor y

H_1 : $(\forall x)((p(x) \wedge \Delta(x)) \rightarrow R(x)) \equiv (\forall x)(\neg p(x) \vee \neg \Delta(x) \vee R(x))$

H_2 : $(\forall x)(v(x) \rightarrow p(x)) \equiv (\forall x)(\neg v(x) \vee p(x))$

H_3 : $(\forall x)(\forall y)((d(x, y) \wedge \neg w(x)) \rightarrow \Delta(x)) \equiv (\forall x)(\forall y)(\neg d(x, y) \vee w(x) \vee \Delta(x))$

H_4 : $(\forall x)(v(x) \rightarrow (\exists y) d(x, y)) \equiv (\forall x)(\neg v(x) \vee (\exists y) d(x, y))$

H_5 : $v(\text{tom}) \wedge \neg w(\text{tom})$

c : $R(\text{tom})$

$H_1^P = (\forall x)(\neg p(x) \vee \neg \Delta(x) \vee R(x))$

$H_2^P = (\forall x)(\neg v(x) \vee p(x))$

$H_3^P = (\forall x)(\forall y)(\neg d(x, y) \vee w(x) \vee \Delta(x))$

$H_4^P = (\forall x)(\exists y)(\neg v(x) \vee d(x, y))$

} premises

$[y \leftarrow f(x)], f(x) = \text{Skolem function}$

$$H_4^S = (\forall x)(\neg w(x) \vee d(x, f(x)))$$

$$C_1 = \neg p(x) \vee \neg \Delta(x) \vee R(x)$$

$$C_2 = \neg w(x) \vee p(x)$$

$$C_3 = \neg d(x, y) \vee w(x) \vee \Delta(x)$$

$$C_4 = \neg w(x) \vee d(x, f(x))$$

$$C_5 = w(\text{dom})$$

$$C_6 = \neg w(\text{dom})$$

$$C_7 = \neg R(\text{dom})$$

$$C_8 = \text{Res}_{\neg w}(C_4, C_5) = d(\text{dom}, f(\text{dom}))$$
$$\theta_1 = [x \leftarrow \text{dom}]$$

$$C_9 = \text{Res}(C_8, C_3) = w(\text{dom}) \vee \Delta(\text{dom})$$
$$\theta_2 = [y \leftarrow f(\text{dom})]$$
$$\theta_1 = [x \leftarrow \text{dom}]$$

$$C_{10} = \text{Res}(C_9, C_6) = \Delta(\text{dom})$$

$$C_{11} = \text{Res}(C_{10}, C_1) = \neg p(\text{dom}) \vee R(\text{dom})$$
$$\theta_1 = [x \leftarrow \text{dom}]$$

$$C_{12} = \text{Res}(C_{11}, C_7) = \neg p(\text{dom})$$

$$C_{13} = \text{Res}(C_{12}, C_2) = \neg w(\text{dom})$$
$$\theta_1 = [x \leftarrow \text{dom}]$$

$$C_{14} = \text{Res}(C_{13}, C_5) = \square$$

- $C_1 \wedge C_2 \wedge \dots \wedge C_{14} \vdash^{\text{Res}} \square \Rightarrow H_1, H_2, H_3, H_4, H_5 \models C$, so the conclusion C can be derived from the set of premises $\{H_1, H_2, H_3, H_4, H_5\}$

④ H_1 : Peter likes any plant that is beautiful.

H_2 : Every greenhouse plant is flourishing or it's not beautiful.

H_3 : The rose is a greenhouse plant.

H_4 : A greenhouse plant must be watered regularly to be flourishing.

H_5 : If a plant is not in the sun, it is not watered regularly.

H_6 : The rose is not in the sun, ~~then Peter does not like the rose~~ C : Peter doesn't like the rose.

- Δ = (domain) the set of all plants and people

$r \in \Delta$ (constant)

(rose)

$P \in \Delta$ (constant)

(Peter)

- Predicate symbols.

$l(x, y) : \Delta \times \Delta \rightarrow \{T, F\}$, $l(x, y) = T$ if x likes y

$b(x) : \Delta \rightarrow \{T, F\}$, $b(x) = T$ if x is a beautiful plant

$g(x) : \Delta \rightarrow \{T, F\}$, $g(x) = T$ if x is a greenhouse plant

$f(x) : \Delta \rightarrow \{T, F\}$, $f(x) = T$ if x is flourishing

$w(x) : \Delta \rightarrow \{T, F\}$, $w(x) = T$ if x is watered regularly

$s(x) : \Delta \rightarrow \{T, F\}$, $s(x) = T$ if x is in the sun

$$H_1: (\forall x)(b(x) \rightarrow l(P, x)) \equiv (\forall x)(\neg b(x) \vee l(P, x))$$

$$H_2: (\forall x)(g(x) \rightarrow (f(x) \vee \neg b(x))) \equiv (\forall x)(\neg g(x) \vee \neg f(x) \vee \neg b(x))$$

$$H_3: g(r)$$

$$H_4: (\forall x)((g(x) \wedge f(x)) \rightarrow w(x)) \equiv (\forall x)(\neg g(x) \vee \neg f(x) \vee w(x))$$

$$H_5: (\forall x)(\neg s(x) \rightarrow \neg w(x)) \equiv (\forall x)(s(x) \vee \neg w(x))$$

$$H_6: \neg s(r)$$

$$C: \neg l(P, r)$$

$$H_5 \vdash_{\text{univ}}^{\text{inst}} \neg \Delta(x) \rightarrow \neg w(x) \quad (\#7)$$

$$H_6, \#7 \vdash_{\text{modus}}^{\text{ponens}} \neg w(x) \quad (\#8)$$

$$H_4 \vdash_{\text{univ}}^{\text{inst}} (g(x) \wedge \neg(x)) \rightarrow w(x) \quad (\#9)$$

$$\#8, \#9 \vdash_{\text{modus}}^{\text{tollens}} \neg(g(x) \wedge \neg(x)) \equiv \neg g(x) \vee \neg \neg(x) \equiv g(x) \rightarrow \neg \neg(x) \quad (\#10)$$

$$H_3, \#10 \vdash_{\text{modus}}^{\text{ponens}} \neg \neg(x) \quad (\#11)$$

$$H_2 \vdash_{\text{univ}}^{\text{inst}} g(x) \rightarrow (\neg(x) \vee \neg b(x)) \quad (\#12)$$

$$\equiv \neg g(x) \vee (\neg(x) \vee \neg b(x))$$

$$H_3, \#12 \vdash_{\text{modus}}^{\text{ponens}} \neg(x) \vee \neg b(x) \equiv \neg \neg(x) \rightarrow \neg b(x) \quad (\#13)$$

$$\#11, \#13 \vdash_{\text{modus}}^{\text{ponens}} \neg b(x) \quad (\#14)$$

$$H_1 \vdash_{\text{univ}}^{\text{inst}} b(x) \rightarrow \Delta(p, x) \equiv \neg \neg \Delta(p, x) \rightarrow \neg b(x) \quad (\#15)$$

$$\equiv \neg \neg \Delta(p, x) \rightarrow \neg b(x)$$

$$\#14, \#15 \vdash_{\text{modus}}^{\text{ponens}} \neg \Delta(p, x)$$

conclusion: The sequence $\{H_1, H_2, \dots, H_6, \#7, \dots, \#15\}$ is the deduction of c from the premises $\{H_1, \dots, H_6\}$.

① Propositional Logic

H₁: To get to work on time, I must take the bus or drive my car.

H₂: If I take the bus, I need to have exact change for the fare.

H₃: If I drive my car, I will have to find a parking spot in the city center.

H₄: I do not have exact change for the fare.

H₅: If I have to find a parking spot, I will be stressed before the meeting starts.

C: If I get to work on time, I will be stressed before the meeting starts.

$$H_1: w \rightarrow b \vee c$$

$$H_2: b \rightarrow e$$

$$H_3: c \rightarrow ps$$

$$H_4: \neg e$$

$$H_5: ps \rightarrow \Delta$$

$$C: w \rightarrow \Delta$$

Legend: w = get to work on time
 b = take the bus
 c = drive my car
 ps = find a parking spot
 Δ = be stressed before the meeting

$$H_1 \wedge H_2 \wedge H_3 \wedge H_4 \wedge H_5 \wedge \neg C \quad (1) \checkmark$$

$$H_1: w \rightarrow b \vee c \quad (2) \checkmark$$

$$H_2: b \rightarrow c \quad (3) \checkmark$$

$$H_3: c \rightarrow ps \quad (4) \checkmark$$

$$H_4: \neg e$$

$$H_5: ps \rightarrow \Delta \quad (5) \checkmark$$

$$\neg C: \neg(w \rightarrow \Delta) \quad (6) \checkmark$$

$$| \alpha \text{ for } (6)$$

$$\neg$$

$$\neg \Delta$$

$$\neg w \quad \beta \text{ for } (2)$$

$$b \vee c \quad (7)$$

$$\otimes = \quad \beta \text{ for } (7)$$

$$\otimes = \quad \beta \text{ for } (3)$$

$$\neg b \quad e \quad \neg c \quad ps \quad \beta \text{ for } (4)$$

$$\otimes = \quad \otimes = \quad \otimes = \quad \beta \text{ for } (5)$$

$$\neg \Delta \quad \Delta \quad \otimes$$

Conclusion

$H_1 \wedge H_2 \wedge H_3 \wedge H_4 \wedge H_5 \wedge \neg C$ has a closed semantic

tableaux $\Rightarrow H_1, H_2, H_3, H_4, H_5 \models C$

② H_1 : If I find a seat near a power outlet and I have my charger, I will stay for three hours.

H_2 : If I stay for three hours, I will finish my essay.

H_3 : I have my charger.

H_4 : I did not finish my essay.

C : I did not find a seat near a power outlet.

$$H_1: (p \wedge c) \rightarrow t$$

$$H_2: t \rightarrow e$$

$$H_3: c$$

$$H_4: \neg e$$

$$C: \neg p$$

$$\text{Legend} \begin{cases} p = \text{I find a seat near an outlet} \\ c = \text{I have my charger} \\ t = \text{I stay for 3 hours} \\ e = \text{I finish my essay} \end{cases}$$

$$H_1 \equiv \neg(p \wedge c) \vee t \equiv \neg p \vee \neg c \vee t \quad (C_1)$$

$$H_2 \equiv t \rightarrow e \equiv \neg t \vee e \quad (C_2)$$

$$H_3 \equiv c \quad (C_3)$$

$$H_4 \equiv \neg e \quad (C_4)$$

$$\neg C \equiv p \quad (C_5)$$

$$\{C_1, C_2, C_3, C_4, C_5\} \vdash_{\text{Res}}^? \square$$

$$C_6 = \text{Res}_p(C_5, C_1) = \neg e \vee t$$

$$C_7 = \text{Res}_e(C_3, C_6) = t$$

$$C_8 = \text{Res}_t(C_7, C_2) = e$$

$$C_9 = \text{Res}_e(C_4, C_8) = \square$$

$$\text{Conclusion: From } \text{CNF}(H_1 \wedge H_2 \wedge H_3 \wedge H_4 \wedge \neg C) \vdash_{\text{Res}} \square \text{ so } C$$

is deducible from the hypotheses

③ H_1 : John buys a pumpkin if and only if he intends to ~~celebrate~~ celebrate the festival.

H_2 : If John intends to celebrate the festival, then he either carves the pumpkin or eats the pumpkin.

H_3 : John does not eat the pumpkin.

H_4 : John bought a pumpkin.

C : John carves the pumpkin.

Legend: $\left\{ \begin{array}{l} b = \text{John buys a pumpkin} \\ f = \text{John celebrates the festival} \\ c = \text{John carves the pumpkin} \\ e = \text{John eats the pumpkin} \end{array} \right.$

$H_1: b \leftrightarrow f$

$H_2: f \rightarrow (c \vee e)$

$H_3: \neg e$

$H_4: b$

$C: c$

We need to prove that:

$$H_1 \wedge H_2 \wedge H_3 \wedge H_4 \models C$$

According to the truth table (next page), ⁱⁿ all the interpretations where all premises are true (interpretation 2), the conclusion is also true. $\Rightarrow$ the conclusion C can be derived from the set of hypotheses $\{H_1, H_2, H_3, H_4\}$

b	$\neg b$	c	$\neg c$	$c \vee \neg c$	$H_1: b \leftrightarrow \neg b$	$H_2: \neg b \rightarrow (c \vee \neg c)$	$H_3: \neg \neg c$	$H_4: b$	$c: c$
T	F	T	F	T	F	T	T	T	T
T	F	T	F	T	F	T	T	T	T
T	F	F	T	T	T	F	F	F	F
T	F	F	T	T	T	F	F	F	F
F	T	T	F	T	T	T	T	T	T
F	T	T	F	T	T	T	T	T	T
F	T	F	T	T	F	F	F	F	F
F	T	F	T	T	F	F	F	F	F

④ H_1 : To enter the vault, the thief must have the fingerprint or the retinal scan.

H_2 : If the thief has the fingerprint, then the security guard was bribed.

H_3 : The security guard was not bribed.

H_4 : If the thief has the retinal scan, then the power must be cut.

H_5 : If the power is cut, the silent alarm will definitely trigger.

H_6 : The thief successfully entered the vault.

C : The silent alarm was triggered.

H_1 : $v \rightarrow f \vee r \equiv v \rightarrow (f \rightarrow r)$	Legend	v - entered the vault
H_2 : $f \rightarrow b$		f - has fingerprint
H_3 : $\neg b$		r - has retinal scan
H_4 : $r \rightarrow p$		b - guard bribed
H_5 : $p \rightarrow a$		p - power is cut
H_6 : v		a - alarm is triggered
C : a		

$H_3, H_2 \vdash_{\text{modus tollens}} \neg f (\neg f)$

$H_6, H_1 \vdash_{\text{modus ponens}} \neg f \rightarrow r (f_8)$

$f_8, \neg f \vdash_{\text{modus ponens}} r (f_9)$

$f_9, H_4 \vdash_{\text{modus ponens}} p (f_{10})$

$f_{10}, H_5 \vdash a$

The sequence $\{H_1, H_2, \dots, H_6, f_8, f_9, f_{10}\}$ is the deduction of C from the set of premises $\{H_1, H_2, \dots, H_6\}$

⑤ H₁: The tracker jacker nest is not disturbed and the fire is colder than yesterday.

H₂: The rebels will attack only if the tracker jacker nest is disturbed.

H₃: If the rebels do not attack, then Katniss will whistle to the mockingjays.

H₄: If Katniss whistles to the mockingjays, then the signal will reach the woods by sunset.

C: The signal will reach the woods by sunset.

$$H_1: \neg \Delta \wedge F$$

$$H_2: A \rightarrow \Delta \equiv \neg A \vee \Delta$$

$$H_3: \neg A \rightarrow W \equiv A \vee W$$

$$H_4: W \rightarrow S \equiv \neg W \vee S$$

$$C: S$$

Legend

- Δ : the tracker jacker nest is disturbed
- F : the fire is colder than yesterday
- A : the rebels attacked
- W : Katniss whistles to the mockingjays
- S : signal reaches the woods by sunset

$$C_1 = \neg \Delta (1)$$

$$C_2 = F (2)$$

$$C_3 = \neg A \vee \Delta (4)$$

$$C_4 = A \vee W (6)$$

$$C_5 = \neg W \vee S (8)$$

$$C_6 = \neg S (9)$$

$$C_7 = \text{Res}_A^{\text{lock}}(C_3, C_4) = \Delta (4) \vee W (6)$$

$$C_8 = \text{Res}_\Delta^{\text{lock}}(C_1, C_7) = W (6)$$

$$C_9 = \text{Res}_W^{\text{lock}}(C_5, C_8) = S (8)$$

$$C_{10} = \text{Res}_S^{\text{lock}}(C_9, C_6) = \square$$

~~H₁ ∧ H₂~~ $C_1 \wedge C_2 \wedge C_3 \wedge C_4 \wedge C_5 \wedge C_6 \vdash_{\text{Res}}^{\text{lock}} \square \rightarrow C$ is deducible from the set of premises $\{H_1, H_2, H_3, H_4\}$